Customer Shopping Behaviour Analysis

(Project Report)

1. Project Overview

This project analyses customer shopping behaviour using transactional retail data to uncover insights related to customer demographics, purchasing patterns, product preferences, discount usage, and subscription behaviour. The objective is to help businesses improve marketing strategies, customer engagement, and revenue optimization through data-driven decision-making.

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2. Problem Statement

A retail company wants to understand customer purchasing behaviour to improve sales performance, customer satisfaction, and long-term loyalty.
Key focus areas include:

- Customer spending patterns
- Impact of discounts and promotions
- Product category performance
- Subscription behaviour
- Demographic-based purchasing trends

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3. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18

Key Features:

- Customer Demographics: Age, Gender, Location, Subscription Status
- Purchase Details: Item Purchased, Category, Purchase Amount, Season, Size, Colour
- Behavioural Attributes: Discount Applied, Review Rating, Frequency of Purchases
- Logistics: Shipping Type

Missing Data:

- Review Rating column had missing values which were handled during preprocessing.

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4. Data Cleaning & Preprocessing (Python)

Data preprocessing was performed using Python (Pandas) to ensure clean and reliable analysis.
Steps Performed:

- Loaded dataset using Pandas
- Checked structure using `df.info()` and `df.describe()`

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31

```python
df.describe()
```

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- Identified missing values using `df.isnull().sum()`

```python
df.isnull().sum()
```

```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

- Handled missing values in Review Rating using **median per category**
- Standardized column names to snake_case format

```python
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

- Removed redundant column (`promo_code_used`)
- Verified data consistency across discount fields

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5. Feature Engineering

New features were created to improve analysis depth:

- **Age Group:** Customers segmented into age categories using quantile-based binning

```
df[["age","age_group"]].head(100)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
...	...	...
95	37	Adult
96	32	Adult
97	21	Young Adult
98	20	Young Adult
99	26	Young Adult

100 rows × 2 columns

- **Purchase Frequency (Days):** Converted categorical frequency into numeric values for analysis

```
df[["frequency_of_purchases", "purchase_frequency_days"]].he
```

	frequency_of_purchases	purchase_frequency_days
0	Fortnightly	14
1	Fortnightly	14
2	Weekly	7
3	Weekly	7
4	Annually	365
5	Weekly	7
6	Quarterly	90
7	Weekly	7
8	Annually	365
9	Quarterly	90

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6. Database Integration (PostgreSQL)

The cleaned dataset was connected to PostgreSQL using SQLAlchemy and loaded into a database table for structured querying.
This enabled:

- Efficient SQL-based business analysis
- Scalable querying of large dataset
- Integration with BI tools

```python
from sqlalchemy import create_engine

# Step 1: Connect to PostgreSQL
username = "postgres"
password = "your_password"
host = "localhost"
port = "5432"
database = "Customer_Behavior_Analysis"

engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

# Step 2: Load DataFrame into PostgreSQL
table_name = "customer"
df.to_sql(table_name, engine, if_exists = "replace", index = False)


print(f"Data successfully loaded into table '{table_name}' in database '{database}'.")
```

```
Data successfully loaded into table 'customer' in database 'Customer_Behavior_Analysis'.
```

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7. SQL Business Analysis

SQL queries were used to extract business insights from customer behaviour data.

Business Problems Solved:

1. Revenue by Gender

Analysed total revenue contribution of male vs female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Customers

Identified customers using discounts but still spending above average.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

3. Top Rated Products

Found products with highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type Analysis

Compared spending behaviour between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscription Analysis

Compared spending patterns of subscribers vs non-subscribers.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount Dependency

Identified products heavily influenced by discount usage.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Customer Segmentation

Grouped customers into:

- New
- Returning
- Loyal

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top Products per Category

Identified top 3 products in each category based on purchase volume.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Buyers Analysis

Checked relationship between repeat buyers and subscription behaviour.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue by Age Group

Calculated revenue contribution across different age segments.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

8. Power BI Dashboard

An interactive dashboard was developed using Power BI to visualize key insights.

Dashboard Highlights:

- Revenue trends by category
- Age group purchasing behaviour
- Subscription vs non-subscription comparison

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9. Key Insights

- The platform has **3.9K customers**, indicating a moderately active user base with stable engagement.
- Average purchase value is **$59.76**, reflecting **mid-range spending behavior**.
- Average customer rating is **3.75**, showing generally satisfied customers but with room for improved experience.
- Only **27% of customers are subscribers**, while **73% are non-subscribers**, highlighting a strong opportunity to improve subscription adoption.
- **Clothing is the top-performing category** in both revenue and sales, followed by accessories, while footwear and outerwear contribute the least, indicating limited category diversification.
- **Young adults are the primary revenue drivers**, followed by middle-aged customers, while adults and seniors contribute minimally.
- Sales and revenue trends are well-aligned across categories and age groups, showing a balanced pricing-to-demand structure.
- Overall customer satisfaction is moderate, suggesting potential improvements in **delivery experience, product quality consistency, and expectation management**.

Business Opportunities

- Increase subscription conversion (currently low at 27%)
- Improve performance of underperforming categories like footwear and outerwear

- Strengthen retention strategies targeting young adults (core customer segment)

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10. Business Recommendations

- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.

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11. Conclusion

This project successfully demonstrates an end-to-end data analytics pipeline from raw data processing to business insights generation.

The insights derived can help retail businesses:

- Improve customer targeting
- Optimize marketing strategies
- Increase revenue efficiency
- Enhance customer satisfaction